

ERIC LIU

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Education

University of Toronto

Sep 2022 – May 2027

BASc, Computer Engineering | Dean's List Scholar | Minors in Artificial Intelligence Engineering, Engineering Business

Experience

Scotiabank | Capital Markets Data Analyst Co-op

May 2026 – Aug 2026

Client Insights & Data Analytics, GBM Data Quality

Toronto, ON

- Built a scalable SQL rule-generation framework that translated critical data elements into **3,000+ data quality checks** for completeness, validity, consistency, and accuracy.

Scotiabank | Quantitative Developer Co-op, Algorithmic Trading

Nov 2025 – Apr 2026

Electronic Execution Services, Global Banking & Markets

Toronto, ON

- Improved VWAP execution performance by **7%** by modeling Canadian and U.S. equity order flow with Gaussian Mixture Models and reinforcement-learning-based analysis, reducing slippage and improving execution-curve stability.
- Built a Bloomberg-integrated outage platform that detected failed algorithmic executions and automated refund and market-impact calculations across **20,000+ orders**.
- Developed Python and SQL analytics using FIX execution data to reconstruct order lifecycles, diagnose routing discrepancies, and investigate fill-quality issues.

Scotiabank | Global Equity Trading Developer Co-op

May 2025 – Oct 2025

Electronic Execution Services, Global Banking & Markets

Toronto, ON

- Developed full-stack trading tools using Python, SQL, Angular, and Bloomberg APIs for execution analytics, client reporting, and post-trade investigations.
- Redesigned commission-calculation workflows for client-specific rates, exchange fees, account rules, and complex trade structures.

Big Tuna | Software Game Developer

Aug 2020 – May 2024

Discord Game Platform, Three-Person Development Team

Oakville, ON

- Co-developed and maintained a JavaScript and SQL game platform serving **25,000+ active players**.

Research

Scotiabank | FX Quantitative Trading Researcher, Incoming

Sep 2026 – Apr 2027

Electronic Execution Services, Global Banking & Markets

Toronto, ON

- Alpha-signal research focused on FX market microstructure, execution modeling, and systematic strategy development.

Research Assistant | Reinforcement Learning for Market Making

May 2023 – Aug 2024

Professor Wei Xu

Toronto, ON

- Developed cryptocurrency market-making agents using DDPG and DQN with reward functions for inventory risk, spread capture, execution, and profitability.
- Built a Python simulation and backtesting environment from **100M+ order-book records** to evaluate trading policies.

Competitions & Leadership

University of Toronto Engineering Competition | *1st Place, Programming Competition*

2023

Ontario Engineering Competition | *University of Toronto Programming Representative*

2024

IMC Prosperity Trading Competition | *6th Place in Canada, Manual Trading*

2026

University of Toronto Aerospace Team | *Developed 6DoF propulsion and aerodynamics simulations*

2023 – 2024

Technical Skills

Programming: Python, C++, C, SQL, Java, JavaScript, TypeScript, Assembly, HTML/CSS

Libraries & Tools: PyTorch, scikit-learn, Pandas, NumPy, Git, Bloomberg API, FIX protocol, Excel, VBA

Trading & Domains: Market microstructure, execution analytics, statistical modeling, machine learning, backtesting